

Continuous Dynamical Systems¹

- † 1. Represent the phase portrait of the scalar dynamical system $\dot{x} = 1 - x^2$. Find $\varphi(t, 1)$ and justify. Specify the properties of the functions $\varphi(t, 2)$ and, respectively, $\varphi(t, 0.5)$.
- † 2. Let $0 < c < 1$ be a parameter and consider the scalar dynamical system $\dot{x} = x(1 - x) - cx$.
- Find its equilibria and study their stability using the linearization method.
 - Represent its phase portrait.
 - When $x(t) > 0$ is considered to be the density of fish in a lake, and $0 < c < 1$ to be the rate of fishing, try to predict the fate of the fish from the lake interpreting the theoretical result obtained at a) and b). $\diamond$
- † 3. Let $c > 1/4$ be a parameter and consider the scalar dynamical system $\dot{x} = x(1 - x) - c$.
- Represent its phase portrait.
 - When $x(t) \geq 0$ is considered to be the density of fish in a lake, and $c > 1/4$ to be the rate of fishing, try to predict the fate of the fish from the lake interpreting the theoretical result obtained at a). $\diamond$
- † 4. Represent the phase portrait of the scalar dynamical system $\dot{x} = 2 - x^2$. Study the stability of the equilibrium points using the linearization method. Find $\varphi(t, \sqrt{2})$ and specify the properties of the functions $\varphi(t, -1.5)$, $\varphi(t, 0)$ and $\varphi(t, 2)$. $\diamond$
- † 5. Represent the phase portrait of the scalar dynamical system $\dot{x} = 2x - x^2$. Study the stability of the equilibrium points using the linearization method. Find $\varphi(t, 2)$, $\varphi(t, 0)$ and study the properties of the functions $\varphi(t, -2)$, $\varphi(t, 1)$ and $\varphi(t, 3)$.

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If there is an attractor, specify its basin of attraction. $\diamond$

+ **6.** Represent the phase portrait, specify the orbits and emphasize the properties of the solution $\varphi(\cdot, \eta)$ for any $\eta \in \mathbb{R}$ for the following equations. Specify the stability of the equilibria.

(a) $\dot{x} = x - x^2$, (b) $\dot{x} = x - x^3 + 1$, (c) $\dot{x} = x - x^3 + 0.2$, (d) $\dot{x} = -x - x^3 + 1$,
(e) $\dot{x} = 2 \sin x$, (f) $\dot{x} = 1 - 2 \sin x$, (g) $\dot{x} = 1 - \sin x$, (h) $\dot{x} = 2 - \sin x$, (i) $\dot{x} = \tanh x$. $\diamond$

+ **7.** Represent the phase portrait of $\dot{x} = \lambda - x^2$. Discuss with respect to the parameter $\lambda \in \mathbb{R}$. $\diamond$

+ **8.** We consider the following linear planar system

(a) $\dot{x} = -6x$, $\dot{y} = -3y$, ~~(b)~~ $\dot{x} = -6y$, $\dot{y} = -3x$, (c) $\dot{x} = 6y$, $\dot{y} = -3x$, (d)
 $\dot{x} = x$, $\dot{y} = -3y$, ~~(e)~~ $\dot{x} = 2y$, $\dot{y} = -3x$, (f) $\dot{x} = -2y$, $\dot{y} = 2x$.

i) Find its flow.

ii) Specify the type and stability of this linear system.

iii) Find a first integral. There is a global first integral?

iv) Represent its phase portrait. $\diamond$

— **9.** Specify the type and stability of the linear systems

~~(a)~~ $\dot{x} = 4x - 5y$, $\dot{y} = x - 2y$, (b) $\dot{x} = x + y$, $\dot{y} = -2x + 4y$, (c)
 $x' = x + y$, $y' = x - 4y$. $\diamond$

+ **10.** (i) For what values of the real parameter a , the system $\dot{x} = ax - 5y$,
 $\dot{y} = x - 2y$ is a center?. In that cases find the general solution of the system.

(ii) For what values of the real parameter a , the system from (i) has a line filled with equilibrium points? $\diamond$

+ **11.** There are uncoupled linear systems which are centers? $\diamond$

12. Find all the equilibrium points of the nonlinear planar system $\dot{x} = x(1 - x)$,
 $\dot{y} = (y + 1)(y - 2)$. $\diamond$

13*. Represent the phase portrait of the following uncoupled (product) systems

(a) $\dot{x} = x(1 - x)$, $\dot{y} = 0$, (b) $\dot{x} = x(1 - x)$, $\dot{y} = y$, (c) $\dot{x} = x(1 - x)$, $\dot{y} = (y + 1)(y - 2)$. $\diamond$

14. Find the equilibrium points and decide whether they are or not hyperbolic, for the nonlinear planar system $\ddot{\theta} + \dot{\theta} + \theta^3 = 0$. $\diamond$

15. We consider the following nonlinear planar systems

$$\dot{x} = -x + xy, \quad \dot{y} = -2y + 3y^2.$$

a) Find its equilibrium points and study their stability using the linearization method.

b) Find $\varphi(t, 0, 2/3)$, $\varphi(t, 4, 0)$ and $\varphi(t, 1, 2/3)$.

c) Represent in the phase plane the orbits corresponding to the initial values $(0, 2/3)$, $(4, 0)$ and $(1, 2/3)$. $\diamond$

16. Consider the following planar system

$$\dot{x} = 2x - x^2, \quad \dot{y} = y - xy^2.$$

a) Find the equilibrium points and study their stability.

b) Find $\varphi(t, 2, 1/2)$, $\varphi(t, 2, 0)$ and $\varphi(t, 0, 2)$. $\diamond$

17. Find the polar coordinates of the following points of cartesian coordinates. Represent all these points in the plane.

$(0, 1)$, $(1, 0)$, $(2, 0)$, $(0, 3)$, $(-3, 0)$, $(0, -2)$, $(1, 1)$, $(1, 1/2)$, $(-2, 1)$, $(-6, -3)$,
 $(\eta_1 \cos t - \eta_2 \sin t, \eta_1 \sin t + \eta_2 \cos t)$ where $t \in \mathbb{R}$ and $(\eta_1, \eta_2) \in \mathbb{R}^2 \setminus \{(0, 0)\}$,
 $(\eta_1 e^t \cos t - \eta_2 e^t \sin t, \eta_1 e^t \sin t + \eta_2 e^t \cos t)$ where $t \in \mathbb{R}$ and $(\eta_1, \eta_2) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.
 $\diamond$

18. We consider the linear planar system $\dot{x} = -x + y$, $\dot{y} = -x - y$. Specify its type and stability. Pass to polar coordinates. Represent its phase portrait. $\diamond$

19. Consider the following planar system

$$\dot{x} = x - 2xy, \quad \dot{y} = x - y.$$

Find the equilibrium points and study their stability. $\diamond$

20. Consider the following planar system

$$\dot{x} = -y(x^2 + y^2), \quad \dot{y} = x(x^2 + y^2).$$

- a) Does this system have other equilibria besides $(0, 0)$? Justify.
- b) Decide whether the equilibrium point $(0, 0)$ is hyperbolic or not.
- c) Verify that $\varphi(t, 1, 0) = (\cos t, \sin t)$, $\varphi(t, 2, 0) = (2 \cos 4t, 2 \sin 4t)$ for all $t \in \mathbb{R}$. Find $\varphi(t, 3, 0)$. Represent the corresponding orbits.
- d) Pass to polar coordinates and represent the phase portrait. Deduce that all the solutions of the system are periodic. $\diamond$

21. We consider the planar system

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2).$$

- a) Study the stability of the equilibrium point $(0, 0)$ using the linearization method. There are other equilibrium points?
- b) Check that $\varphi(t, 1, 0) = (\cos t, \sin t)$ for all $t \in \mathbb{R}$. Represent the corresponding orbit. $\diamond$

22. Consider the following planar system

$$\dot{x} = -y - x(x^2 + y^2), \quad \dot{y} = x - y(x^2 + y^2).$$

- a) Does this system have other equilibria besides $(0, 0)$? Justify.
- b) Prove that the equilibrium point $(0, 0)$ is not hyperbolic.
- c) Use the system obtained by passing to polar coordinates to determine the shape of the orbits. $\diamond$

23. Denote by $x(t)$ the density of a trout population at time t . Describe its evolution in each of the following cases.

- (a) $\dot{x} = 100x$.
- (b) $\dot{x} = 100x - x^2$.

Assume that someone is fishing, first with a constant rate $k > 0$, that is,

- (c) $\dot{x} = 100x - x^2 - k$,

and another one, in another basin, with rate proportional to the density, $kx(t)$, where $k > 0$, that is,

- (d) $\dot{x} = 100x - x^2 - kx$.

Discuss with respect to k . $\diamond$

24. For each $k > 0$ we consider the differential equation

$$\dot{x} = -k(x - 21),$$

which is the model of Newton for cooling processes, here $x(t)$ being the temperature of a cup of tea at time t .

(a) Find its flow. (b) An experiment revealed the following fact. A cup of tea with initial temperature of $49^\circ C$ has a temperature of $37^\circ C$ after 10 minutes. Find the initial temperature of a cup of tea such that after 20 minutes the tea has $37^\circ C$. $\diamond$

— **25.** Find a global first integral of the following nonlinear planar system $\ddot{\theta} + \omega^2 \sin \theta = 0$ (where $\omega > 0$). $\diamond$

26. Find a first integral in $(0, \infty) \times (0, \infty)$ of the following nonlinear planar system $\dot{x} = N_1x - xy$, $\dot{y} = -N_2y + xy$ (where $N_1, N_2 > 0$). $\diamond$

27. Find a first integral in $(0, \infty) \times (0, \infty)$ of the following nonlinear planar system $\dot{x} = x - xy$, $\dot{y} = -0.3y + 0.3xy$. $\diamond$

— **28.** a) Give an example of a coupled linear planar system which has a node at the origin.

b) Give an example of a coupled linear planar system which has a saddle at the origin. $\diamond$

— **29.** Find the the flow of each of the systems. Decide whether, for every $\eta \in \mathbb{R}^2$ we have $\lim_{t \rightarrow \infty} \varphi(t, \eta) = 0$ or we have that $\varphi(t, \eta)$ is bounded for $t \in (0, \infty)$.

a) $x' = -2x$, $y' = -3y$

b) $x' = -2x$, $y' = 3y$

c) $x' = -3x$, $y' = x - 3y$

d) $x' = y$, $y' = \omega^2 x$ (here $\omega > 0$ is a parameter)

e) $x' = -x - y$, $y' = x - y$

f) $x' = -5x - 9y$, $y' = 2x + y$. $\diamond$